

Hiera Perceptual Loss on Frequency Domain

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Abstract

We propose a novel loss function for image super-resolution (SR) that integrates hierarchical perceptual features with frequency domain supervision. Specifically, our approach leverages a hierarchical perceptual model (Hiera) to extract deep, multi-scale feature representations from both low-resolution inputs and high-resolution ground truths. A frequency loss is computed on these perceptual features after transformation into the frequency domain, capturing discrepancies in both amplitude and phase. This frequency loss is further combined with a traditional mean-squared-error (MSE) loss in the pixel space to form an overall training objective that balances global structure preservation with fine-detail recovery. This result suggests that while frequency-based supervision affects the perceptual feature domain, further refinements may be needed to achieve significant metric gains.¹

1. Introduction

Image super-resolution (SR) is a fundamental task in computer vision with broad applications in medical imaging, remote sensing, surveillance, and consumer photography. By reconstructing high-resolution (HR) images from low-resolution (LR) inputs, SR enables enhanced visual quality and improved performance in downstream tasks such as detection and recognition. However, due to the inherently ill-posed nature of the problem, effective SR remains particularly challenging and requires carefully designed optimization strategies [7].

Traditional SR methods typically minimize per-pixel losses such as mean squared error (MSE), leading to high pixel fidelity but often resulting in overly smooth images that lack realistic textures. To address this, perceptual losses—based on high-level features extracted by deep convolutional neural networks—have gained popularity for producing visually pleasing outputs [10]. These losses cap-

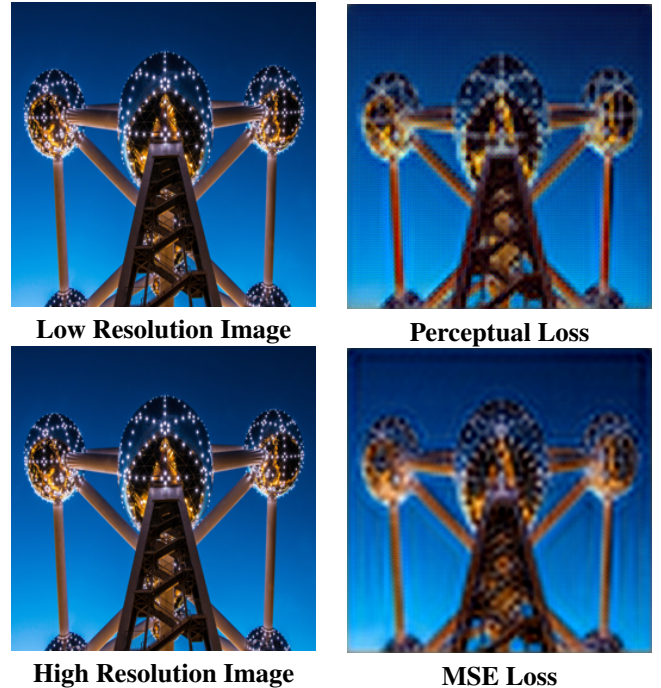


Figure 1: Comparison of different loss methods on Super Resolution.

ture structural and semantic characteristics beyond pixel-wise similarity.

A central challenge in SR lies in balancing pixel fidelity and perceptual quality. While the former ensures numerical similarity to the ground truth, it often sacrifices sharpness. The latter focuses on preserving meaningful structures such as edges and textures, which are crucial for visual realism and details [17, 2].

Recent advances in computer vision, particularly transformer-based models, have enabled better modeling of long-range dependencies and global context. Vision Transformers (ViT) [5] and hierarchical transformer architectures provide a powerful attention-based mechanism for aggregating multi-scale information. Such models are especially

¹GitHub: https://github.com/EnigmaticAbyss/super_resolution_loss

effective in SR, where capturing both global structure and fine detail is essential.

Building on these developments, we propose the usage of a hierarchical perceptual model that replaces traditional VGG-based loss computation with a transformer-based architecture. To further improve reconstruction quality, we introduce a hierarchical perceptual loss that captures image features across multiple levels of abstraction [15]:

Shallow layers emphasize high-frequency details such as textures and edges.

Mid-level layers extract semantic features that connect local textures to global shapes.

Deep layers capture global structures, contextual coherence, and information.

By leveraging multi-scale perceptual cues from different layers, our method achieves super-resolved outputs that are both rich in fine detail and globally consistent.

2. Related Work

Super-resolution research has advanced along multiple fronts, particularly in the design of loss functions, frequency-aware supervision, and hierarchical architectures. We review representative approaches in each area, which collectively inform the development of our method.

2.1. Pixel-Wise and Perceptual Losses

Early deep learning-based SR methods, such as SRCNN [4], primarily optimized for pixel-wise losses like Mean Squared Error (MSE). While effective at maximizing numerical fidelity, these approaches often produced overly smooth outputs that lack high-frequency detail, due to their limited sensitivity to perceptual attributes.

To address this shortcoming, perceptual loss functions were introduced by Johnson *et al.* [10], leveraging feature activations from pre-trained networks like VGG [18] to guide training. These losses encourage perceptual similarity by aligning semantic structures between images, significantly improving visual realism. However, their reliance on fixed feature extractors restricts adaptability, particularly in restoring intricate textures.

Subsequent approaches have proposed hybrid objectives that combine pixel-wise and perceptual cues, as well as adaptive formulations that prioritize difficult regions or emphasize multi-scale consistency [9]. These strategies offer improved trade-offs between distortion metrics and perceptual quality.

2.2. Frequency Domain Supervision

Incorporating frequency-domain information has emerged as a complementary strategy to enhance SR performance. Jiang *et al.* [9] introduced Focal Frequency Loss, which adaptively weighs frequency components based on

reconstruction difficulty, directing attention toward regions with high perceptual impact. Similarly, Fuoli *et al.* [6] proposed a Fourier-domain loss to supervise the generator directly in restoring missing spectral components, thereby enriching high-frequency details.

Sims [19] further explored frequency-guided perceptual loss, informed by principles from image compression (e.g., JPEG quantization), to enhance texture recovery. Collectively, these studies underscore the benefit of frequency-aware supervision as a means of complementing spatial losses, particularly for reconstructing fine-grained structures that are perceptually salient but difficult to recover using spatial cues alone.

2.3. Hierarchical Architectures

Hierarchical models have gained traction for their ability to learn and represent features across multiple spatial resolutions. One notable example is Hiera [16], introduced by Facebook Research, which is specifically designed for hierarchical visual understanding and synthesis. By progressively aggregating features at different scales, Hiera captures both global context and local detail more effectively than shallow or single-resolution networks.

Our approach builds on this hierarchical paradigm by incorporating Hiera-derived features into the supervision pipeline. In particular, we leverage its multi-scale representations from both ground truth and generated images to guide frequency-aware loss computation. This integration allows the model to refine high-frequency content while maintaining semantic and structural integrity across scales, ultimately improving perceptual quality in SR outputs.

3. Proposed Method

We introduce our approach for image super-resolution (SR). The method is structured around three key components:

First, we apply a frequency loss to perceptual features extracted from deep network layers. This enhances the high-frequency details in the generated image. Second, a traditional MSE loss is employed to ensure pixel fidelity and numerical accuracy in the reconstructed image. Finally, we combine both the frequency and MSE losses into a single training objective, balancing perceptual quality and pixel-level accuracy.

3.1. Frequency Loss on Perceptual Features

After extracting the perceptual feature maps from the chosen layer $\phi_k(\cdot)$, we transform these maps into the frequency domain using the Fast Fourier Transform (FFT).

We define the Fourier transform of the perceptual feature map $\phi_k(x)$ as:

$$F_{u,v}(x) = \mathcal{F}(\phi_k(x)),$$

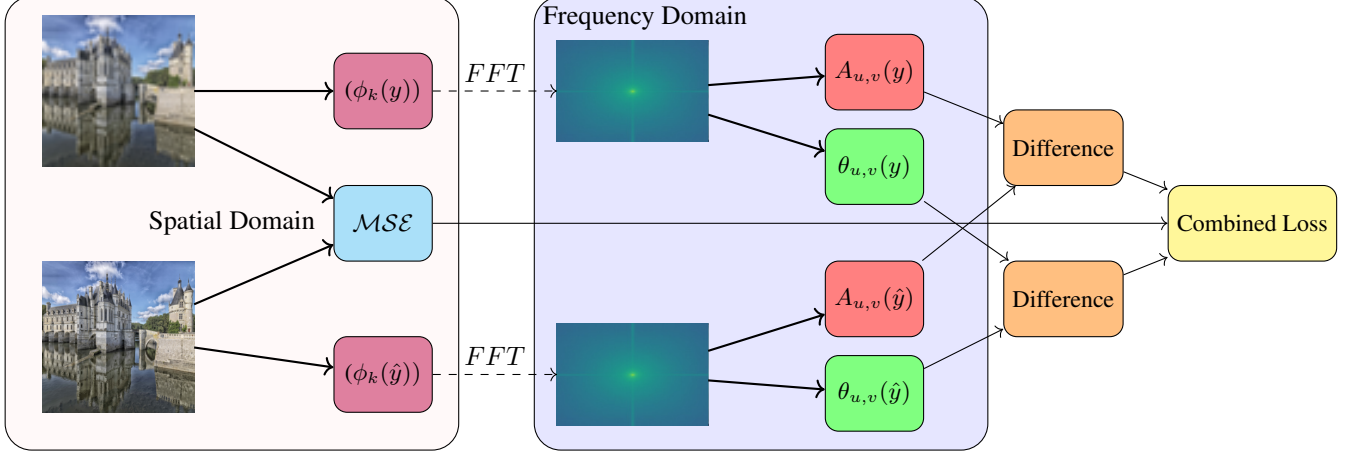


Figure 2: Pipeline of the proposed method.

where $\mathcal{F}$ represents the Fourier transform operator, and $x \in \{y, \hat{y}\}$ represents the ground truth image y or the generated image $\hat{y}$. Here, u and v represent the spatial frequency coordinates in the frequency domain.

We decompose the complex frequency representation into its magnitude (amplitude) and phase components using the following formulas:

Magnitude:

$$A_{u,v}(y) = \sqrt{\Re\{F_{u,v}(y)\}^2 + \Im\{F_{u,v}(y)\}^2}, \quad (1)$$

$$A_{u,v}(\hat{y}) = \sqrt{\Re\{F_{u,v}(\hat{y})\}^2 + \Im\{F_{u,v}(\hat{y})\}^2}, \quad (2)$$

Phase:

$$\theta_{u,v}(y) = \text{atan2}(\Im\{F_{u,v}(y)\}, \Re\{F_{u,v}(y)\}), \quad (3)$$

$$\theta_{u,v}(\hat{y}) = \text{atan2}(\Im\{F_{u,v}(\hat{y})\}, \Re\{F_{u,v}(\hat{y})\}), \quad (4)$$

where $\Re\{\cdot\}$ denotes the real part, and $\Im\{\cdot\}$ denotes the imaginary part of the complex frequency components.

We now compute the frequency loss using two components: one for the amplitude and another for the phase. The dynamic weighting functions $w_A(u, v)$ and $w_\theta(u, v)$ are defined as functions of the absolute differences between the corresponding components of the ground truth and generated feature maps. The amplitude and phase losses are calculated as follows:

$$L_{\text{amp}} = \frac{2}{UV} \sum_{u=0}^{\frac{U}{2}-1} \sum_{v=0}^{V-1} w_A(u, v) \cdot |A_{u,v}(\hat{y}) - A_{u,v}(y)|, \quad (5)$$

$$L_{\text{phase}} = \frac{2}{UV} \sum_{u=0}^{\frac{U}{2}-1} \sum_{v=0}^{V-1} w_\theta(u, v) \cdot |\theta_{u,v}(\hat{y}) - \theta_{u,v}(y)|, \quad (6)$$

where $U \times V$ is the total number of frequency coordinates, and $w_A(u, v)$ and $w_\theta(u, v)$ are the dynamic weights applied to the amplitude and phase components, respectively. These weights are defined as follows:

$$w_A(u, v) = |A_{u,v}(y) - A_{u,v}(\hat{y})|^{\alpha_A}, \quad (7)$$

$$w_\theta(u, v) = |\theta_{u,v}(y) - \theta_{u,v}(\hat{y})|^{\alpha_\theta}, \quad (8)$$

where $\alpha_A, \alpha_\theta > 0$ control the emphasis on frequency components with larger discrepancies.

Finally, the overall frequency loss is given by the sum of the amplitude and phase losses:

$$L_{\mathcal{F}} = \frac{1}{2}L_{\text{amp}} + \frac{1}{2}L_{\text{phase}}. \quad (9)$$

This formulation ensures that both the energy distribution (through the amplitude) and the structural alignment (through the phase) of the feature maps in the frequency domain are matched between the generated and ground-truth images. As a result, the super-resolved image is guided to preserve both global structure and high-frequency details, which are crucial for perceptual quality.

3.2. Mean-Squared-Error Loss in Pixel Space

To ensure overall reconstruction fidelity, we include a standard MSE loss:

$$L_{\text{MSE}} = \frac{1}{HW} \sum_{h=1}^H \sum_{w=1}^W (\hat{y}(h, w) - y(h, w))^2, \quad (10)$$

where H and W denote the image dimensions.

Config	Generator	$\mathcal{L}_{MSE}$	$\mathcal{L}_{Hiera}$	$\mathcal{L}_{PerceptualVGG}$	$\mathcal{L}_{Frequency}$	$\uparrow$ PSNR	$\uparrow$ SSIM	$\downarrow$ LPIPS
1	EDSR [12]		✓			9.7337	0.0143	0.8397
2	EDSR	✓	✓		✓	15.4226	0.2075	0.5943
3	EDSR		✓		✓	8.5721	-0.0125	0.6112
4	EDSR	✓				20.9091	0.6062	0.3805
5	EDSR			✓		19.2594	0.4291	0.252
6	NinaSR [8]		✓			9.7909	0.0153	0.8402
7	NinaSR	✓	✓			18.8545	0.4191	0.2715
9	NinaSR	✓	✓		✓	11.6367	0.0593	0.6102
10	NinaSR	✓				19.9522	0.4922	0.3932
11	NinaSR			✓		16.1827	0.2963	0.2379
12	SwinIR [11]		✓			10.2221	0.0118	0.7603
13	SwinIR	✓	✓		✓	15.9004	0.1861	0.6474
14	SwinIR		✓		✓	11.2945	0.0399	0.6261
15	SwinIR	✓				19.4056	0.3809	0.4657
16	SwinIR			✓		19.6895	0.4773	0.2492

Table 1: Comparison of different configurations.

3.3. Combined Loss Function

The total training loss is defined as:

$$L_{\text{total}} = \lambda_{\text{MSE}} \cdot L_{\text{MSE}} + \lambda_{\text{freq}} \cdot L_{\mathcal{F}}, \quad (11)$$

where λ_{MSE} and λ_{freq} are hyperparameters that balance the contributions of the two losses. Our SR network is trained using gradient descent with these combined losses, with the objective of preserving global structure while enhancing fine high-frequency detail.

4. Experiments and Results

This section presents the evaluation of our method. We first outline the experimental setup, including dataset and training details. Then, we report quantitative results using standard metrics and analyze the impact of different loss functions. Finally, we assess the runtime efficiency of our approach compared to existing methods.

4.1. Experimental Setup

We conducted our experiments on the DIV2K dataset[1], a widely used benchmark for image super-resolution tasks. The DIV2K dataset consists of high-resolution (HR) images, which were downsampled to create low-resolution (LR) counterparts. These HR and LR pairs were used for training and evaluation.

The network was trained with 16-image batches for 100 epochs, utilizing the AdamW optimizer[14] with a carefully tuned learning rate schedule. The optimizer and learning rate schedule were selected to ensure efficient convergence during training and better generalization.

For feature extraction, through cross-validation, we utilized the second Hiera block. We found this block to be particularly effective in extracting relevant features for super-resolution tasks, as it strikes a balance between capturing fine-grained details and global structural information—both of which are critical for high-quality super-resolution.

The model was trained using a combination of perceptual loss and frequency domain loss, as outlined in previous sections. To improve training stability and model convergence, we employed a cosine annealing scheduler for dynamically adjusting the learning rate over time. This approach allowed for more efficient training and ensured better generalization.

Hyperparameters, including the learning rates and the weights for perceptual and frequency losses, were optimized via cross-validation using a validation set from the DIV2K dataset.

The experiments were carried out on a high-performance computing system equipped with NVIDIA A100 GPUs (40 GB memory). The system also featured 2 x AMD EPYC 7662 processors (each with 128 cores running at 2.0 GHz) and a substantial 512 GB of RAM. Additionally, the machine was equipped with 5.8 TB of node-local SSD storage, which provided the necessary resources to handle the large datasets and demanding computations required for our super-resolution experiments.

4.2. Quantitative Evaluation

We assessed our method’s performance using three widely adopted metrics: PSNR, SSIM, and LPIPS.

PSNR (Peak Signal-to-Noise Ratio) evaluates image fidelity by comparing pixel-wise differences. Higher PSNR values indicate better reconstruction quality, with scores

above 30 dB typically considered good.

SSIM (Structural Similarity Index) measures structural similarity between the generated and ground truth images. Values closer to 1 denote high similarity, with scores above 0.9 considered excellent.

LPIPS (Learned Perceptual Image Patch Similarity) quantifies perceptual similarity using deep network features. Lower LPIPS scores indicate that the generated image is perceptually closer to the reference.

Together, these metrics provide a balanced evaluation of both pixel accuracy and perceptual quality.

Table 1 demonstrates that incorporating perceptual and frequency losses enhances image quality. PSNR values increase with the addition of $\mathcal{L}_{MSE}$, $\mathcal{L}_{Hiera}$, and perceptual loss with VGG, with EDSR achieving the highest PSNR (20.9091 dB). Similarly, SSIM improves, especially with Perceptual Loss using VGG, where EDSR also reaches the highest SSIM score (0.6062). LPIPS scores decrease with perceptual losses, with NinaSR showing the best LPIPS (0.2379) when perceptual Loss with VGG is being used. These results suggest that while perceptual and frequency losses enhance structural similarity and perceptual quality, they may trade off PSNR. Among the models, EDSR excels in PSNR and SSIM, while SwinIR performs well in SSIM but has lower PSNR, indicating better structure preservation at the cost of fidelity.

4.3. Qualitative Analysis

Figure 3 presents side-by-side comparisons of the ground truth and outputs from our model alongside baseline methods. While traditional models tend to smooth out high-frequency content, our approach preserves textures and edges, resulting in more visually appealing outputs. This improvement is largely attributed to the combination of perceptual and frequency domain losses, which together help the network capture both local textures and global structures.

To assess the contribution of each component in our framework, we conducted ablation studies using various loss function combinations:

VGG Perceptual Loss evaluates the impact of high-level feature alignment on perceptual quality.

Frequency Loss assesses the contribution of frequency-domain supervision for high-frequency detail recovery.

MSE Loss serves as a baseline to gauge pixel-level fidelity.

MSE + VGG Perceptual Loss balances pixel accuracy with perceptual quality.

Hiera Perceptual uses perceptual features from the second Hiera block to enhance textures and structural consistency without relying on MSE.

Frequency Loss on Hiera Perceptual Features applies frequency loss to perceptual feature maps to improve high-frequency recovery within the feature space.

Hiera + Frequency + MSE Loss through adding MSE to the previous configuration evaluate the influence of pixel-level guidance.

Hiera Perceptual Loss on Magnitude and Phase Differences aligns differences in magnitude and phase between ground truth and generated features to enhance perceptual relevance in the frequency domain.

We tested these configurations using three state-of-the-art super-resolution models:

NinaSR a convolutional architecture with non-local attention that effectively recovers texture details [3].

SwinIR a transformer-based model leveraging the Swin Transformer backbone to capture both local and global dependencies [11, 13].

EDSR a deep residual network designed to enhance reconstruction via deep layers and residual learning [12].

Results in Table 1 show that both perceptual and frequency losses substantially improve visual quality. Excluding either leads to perceptible degradation, particularly in texture fidelity and structural coherence. The combination of hierarchical perceptual and frequency losses yielded the best perceptual results in the Hiera based experiments, while incorporating MSE provided marginal improvements in PSNR and SSIM.

Overall, the ablation study highlights the importance of balancing perceptual and pixel-wise losses to achieve high-quality, visually convincing super-resolution results.

4.4. Runtime and Efficiency

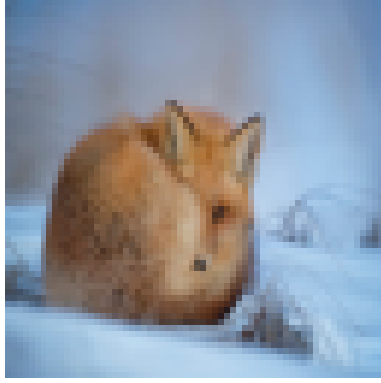
Our method was implemented using a lightweight SR architecture designed to be efficient for real-time applications. The network was trained with the same training time as state-of-the-art methods, and the inference speed is comparable to that of conventional SR networks. This efficiency makes our approach suitable for deployment in real-time scenarios where both quality and speed are critical.

5. Discussion

Our experiments reveal key findings regarding the use of perceptual and frequency loss for image super-resolution.

First, applying frequency loss on perceptual features did not improve results compared to using the VGG perceptual loss alone. Although we hypothesized that frequency-domain supervision would aid in fine detail recovery, the VGG perceptual loss consistently outperformed it in perceptual quality. The VGG loss, with its focus on high-level feature matching, proved more effective in preserving texture and structural integrity.

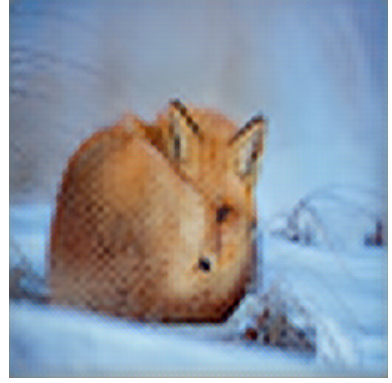
Similarly, Hiera perceptual loss, despite its multi-scale feature extraction, did not surpass VGG perceptual loss. While the hierarchical model captures both fine details and global structures, the VGG loss, tailored for perceptual similarity, delivered superior results in super-resolution tasks.



Low Resolution Image



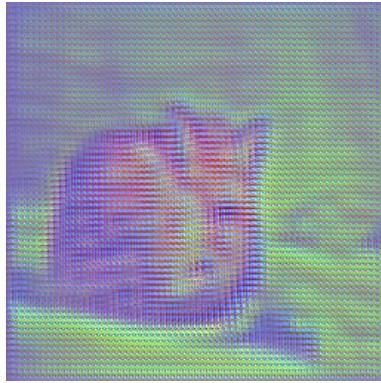
High Resolution Image



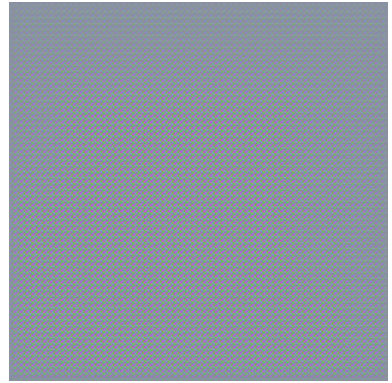
Perceptual Loss



**Hiera Perceptual Loss with
Frequency and MSE Loss**



**Hiera Perceptual Loss with
Frequency and No MSE Loss**



**Hiera Perceptual Loss with No
Frequency Loss and No MSE Loss**

Figure 3: Comparison of Super-Resolution Losses using SwinIR model.

This suggests that the VGG loss is better suited for capturing high-level features.

Interestingly, adding frequency loss to Hiera perceptual loss as shown in Figure 3 improved image quality but still did not match the performance of VGG perceptual loss. The frequency loss enhanced fine detail preservation, but the overall perceptual quality was not as high as VGG, confirming that VGG is more effective for generating high-quality images.

6. Conclusion and Future Work

We have presented a novel loss function for image super-resolution that fuses hierarchical perceptual feature extraction with frequency domain supervision and traditional pixel-wise MSE loss. By transforming multi-scale perceptual features into the frequency domain and applying a dynamic frequency loss, our approach aims to enhance the recovery of high-frequency details while preserving global structure. Although the introduced frequency loss alters the characteristics of the perceptual loss and improves high-frequency modeling, the overall gains in standard quanti-

tative metrics are modest.

Future research directions include:

1. Refining the dynamic weighting mechanism to better balance the contributions of each loss component.
2. Integrating adversarial training in the frequency domain to further boost perceptual quality.
3. Extending the framework to other image restoration tasks, such as denoising or deblurring.
4. Finding a scientific explanation behind the decrease in quality due to the addition of Frequency Loss to Hiera perceptual loss in the frequency domain.

We believe further exploration in these areas will bridge the gap between perceptual improvements and quantitative performance metrics.

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